

Lake Guardians
Design of a Device to Remove Plastics from Lakes in Austin
Final Design Report

4/28/2026

Executive Summary

Plastic pollution in Central Texas lakes is a growing environmental and public health concern, with macroplastics fragmenting into persistent microplastics that accumulate in shoreline sediments. The City of Austin Watershed Protection department conducts daily cleanup of Lady Bird Lake, but their boats draft 12 to 18 inches and cannot easily access the shallow, vegetated brush where debris collects most heavily. Crews currently reach these areas by grabbing from a distance or wading in on foot, bag by bag.

The SS Guardian addresses this gap. It is a small, remote-controlled airboat with a sub-1-inch hull draft, designed to clear shoreline plastic where standard boats cannot reach. A front-mounted, single-piece TPU rake captures floating debris and flexible vegetation, while air propulsion eliminates submerged drivetrain components that would foul in weeds. Sized for use by Watershed crews and volunteer groups, it extends manual cleanup rather than replacing larger interceptor systems.

Concept selection drew on stakeholder interviews with UT researchers, lake users, and Watershed Protection, followed by structured ideation and a Pugh matrix that ranked the airboat highest on maneuverability, safety, capacity, cost, and durability. The design progressed from hand sketch to foam prototype to final test boat, refined through four major revisions: a redesigned motor mount, a one-piece TPU rake, an air rudder in the prop wash, and a foam-safe epoxy and marine topcoat system. The final build is a 24-inch XPS foam hull with a brushed motor, 10-inch nylon prop, and Spektrum RC electronics. On-water testing at Keep Austin Beautiful Day on April 18, 2026, confirmed performance against design targets at \$188 of \$250 budget spent.

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1. Introduction

Plastic pollution in Central Texas lakes remains a persistent and increasingly visible environmental challenge. Recent research at the University of Texas at Austin has shown rising microplastic concentrations in lake sediments, with many particles originating from the breakdown of larger floating plastics over time. These macroplastics accumulate most heavily along brush-filled, shallow shoreline zones where conventional cleanup boats have limited maneuverability, and crews must rely on manual labor methods. Our project responds to this gap by developing a small-scale, deployable cleanup system that can reach areas inaccessible to current technologies while reducing the long-term generation of microplastics. Building on the conceptual design work from the previous semester, the project has now progressed through critical design review and into manufacturing and testing. The design emphasizes affordability, portability, and minimal disturbance to aquatic life and sediment while cooperating with existing cleanup methods.

Overall, this report presents the complete work of the Lake Guardians team in designing, manufacturing, testing, and validating the SS Guardian, a remote-controlled airboat developed to support macroplastic debris collection in Central Texas waterways. Through environmental research, stakeholder engagement, iterative prototyping, hands-on fabrication, and on-water testing, the team has delivered a practical, accessible, and ecologically responsible device that extends the reach of existing watershed protection efforts and helps address the upstream source of microplastic pollution in Austin's lakes.

2. Problem Statement

We aim to design and build a safe, affordable, and field-deployable device to actively remove floating plastic debris from Central Texas lakes, mainly Lady Bird Lake and Lake Austin. The device targets macroplastics introduced through urban runoff, wind and storm, and tire fragments washed through creeks and storm drains. Unlike current cleanup methods that rely heavily on manual labor, our project focuses on removing floating plastics from hard-to-reach and obstructed areas.

Through extensive design research conducted during the concept development phase, including stakeholder interviews with UT researchers, lake users, and representatives from the City of Austin Watershed Protection Department, we identified key requirements such as maneuverability in constrained spaces, resistance to entanglement with vegetation, and safe deployment from shore. These insights drove the use of specified design methodologies, including functional decomposition, morphological matrices, ideation techniques (6-3-5, SCAMPER, and design-by-analogy), and Pugh chart comparisons. This process led the team to select a remotely operated airboat equipped with a front-mounted rake as the final system design, a choice that meets customer needs and engineering requirements while improving upon existing cleanup approaches.

3. Design Research and Alternatives

To inform the selection of design alternatives for a small-scale, deployable cleanup system targeting macroplastic debris in Central Texas lakes, comprehensive research was conducted encompassing market analysis, bio-inspired design analogies, and structured ideation methods. This was augmented by theoretical and numerical

analyses and low-fidelity prototyping to validate feasibility. The following subsections detail each phase of this research.

3.1 Market Analysis

Market research highlighted that plastic pollution in urban lakes like Lady Bird Lake primarily arises from stormwater runoff, carrying macroplastics such as bottles and bags and contributing to microplastic generation through fragmentation. UT Austin studies documented elevated sediment concentrations near inflows, with tire-wear particles predominant [3–5, 8]. Existing solutions span passive interceptors to autonomous vessels, but gaps persist in maneuverability for vegetated shorelines. For instance, the Seabin employs a submersible pump for marina debris capture [10], while the Ocean Cleanup Interceptor uses solar-powered booms and conveyors for river-scale removal [13]. Autonomous options include the WasteShark and Clearbot [14, 17]. Biomimetic designs draw from nature: the Great Bubble Barrier creates upward currents via perforated tubes [15], and Virginia Tech’s manta-ray-inspired array uses flexible polymer gill rakers for clog-resistant filtration [21]. Patents reflect innovation trends, such as US Patent No. 5,499,600 for conveyor skimmers and WO 2012/074240 A2 for passive flow-assisted collectors [23, 24]. However, mid-scale lake applications remain underserved; industrial systems like Mr. Trash Wheel are oversized for shallow zones [9], and static traps like StormX lack adaptability to dynamic hotspots [11].

3.2 Concept Generation

Concept generation employed 5-3-4 brainwriting, with five team members each contributing three ideas in four-minute rounds, yielding a couple dozen concepts through iterative passing and building. Additional methods included SCAMPER, which modified Mr. Trash Wheel for portability by substituting hydropower with batteries, combining with modular arms, and reversing conveyor to floating combs, and Design-by-Analogy, such as baleen whales for filtering, lobster traps for one-way retention, and beaver dams for flow deflection. Five concept variants emerged from this process: a Floating Channel (buoyant passive system with guided surface flow, internal fans, and one-way filter), an Air Boat (remote-controlled surface vessel powered by air propeller with front-mounted rake and wipers), an Autonomous Clean-Up Drone (multi-rotor aerial skimmer with downward-facing cameras and servo-driven claws or nets), a 3-in-1 Trash Cleaning Tool or Net Gun variant (manually operated extendable tool with claw grabber, puncture probe, and retractable net), and a Creek Crawler (wheeled semi-autonomous rover with front rake for shallow inflows). All 5 concept sketches can be found at the top of the Pugh chart in Appendix C.

3.3 Theoretical and Numerical Analyses

Theoretical analyses were performed throughout the project to evaluate concept feasibility, compare alternatives, and validate design decisions on a quantitative basis. These calculations provided first-order estimates of key performance parameters before committing to detailed design and were later compared against measured results during on-water testing.

3.3.1 Propulsion and Drag

Propulsion analysis was used to confirm that the selected motor and propeller combination could overcome the hydrodynamic drag of the hull at cruise speed. Static thrust was estimated using the actuator disk model:

$$F_{thrust} = \rho A v^2 \text{ (Equation 1)}$$

where $\rho = 1.225 \text{ kg/m}^3$ is the density of air, $A = 0.0506 \text{ m}^2$ is the swept area of the 10-inch propeller disk, and $v = 25 \text{ m/s}$ is the estimated exit velocity through the prop. This yielded a predicted thrust of approximately **38.75 N**.

Hull drag at a 1 m/s cruise speed was estimated using the standard drag equation:

$$F_{drag} = \frac{1}{2} \rho A v^2 C_D \text{ (Equation 2)}$$

where $\rho = 1000 \text{ kg/m}^3$ is the density of water, $A = 0.12 \text{ m}^2$ is the estimated wetted area of the hull, and $C_D = 0.2$ is a conservative drag coefficient for a flat-bottomed hull. This produced an estimated drag of approximately **12 N**, yielding a thrust-to-drag margin of roughly **3.25x**, confirming the propulsion system was sized with adequate headroom for pushing through aquatic vegetation.

3.3.2 Buoyancy and Payload Capacity

Hull buoyancy was calculated to ensure adequate freeboard with all components installed. The hull consists of three layers of 25 psi XPS foam encapsulated in a fiberglass and epoxy laminate, with a footprint of approximately 24 in × 12 in and a total thickness of 3 in. Maximum buoyant force was estimated using Archimedes' principle:

$$F_b = \rho_{water} V_{displaced} \text{ (Equation 3)}$$

With a footprint of 2 ft² and 3 in of hull depth, the fully submerged displaced volume is approximately 0.5 ft³, yielding a theoretical maximum buoyant force of approximately **31 lbs**. The measured operating draft under fully loaded conditions was only 0.5 in, indicating that the hull operates well below its buoyancy limit and provides substantial freeboard margin for stability and splash resistance during use.

3.3.3 Debris Interception

Debris interception capacity was estimated using the continuity equation to determine the volumetric flow rate of water passing through the rake intake during operation:

$$Q = A \cdot v \text{ (Equation 4)}$$

For an effective rake width of 0.3 m and a submerged depth of 0.05 m at the measured cruise speed of 1.2 m/s, the theoretical throughput is approximately 1.08 m³/min. The TPU rake was designed to allow water and small vegetation to pass through while retaining floating macroplastic debris, addressing the upstream source of microplastic pollution. Operating speeds were kept below the EPA intake velocity guideline of 0.3 m/s in regions of close approach to the rake to minimize wildlife entrapment risk [15].

3.3.4 Cost Analysis

The project operated under a total budget of \$250. Final out-of-pocket spending totaled **\$188**, with the remaining costs offset by reclaimed XPS foam from a local construction site and electronics (motor, ESC, receiver, battery, servo) provided by The Cockrell School of Engineering. The total replication cost, representing what an external party would pay to reproduce the SS Guardian, is **\$364**, broken down by subsystem as follows:

Subsystem	Cost
Propulsion	\$111
Hull	\$98
Electronics	\$75
Hardware	\$49
3D Printed Components	\$31
Total	\$364

Table 1: Subsystem Cost Breakdown for the SS Guardian

This breakdown demonstrates that the SS Guardian remains a low-cost, accessible platform that could be reproduced by other watershed protection groups or community organizations using widely available materials and standard manufacturing methods.

3.4 Concept Evaluation and Selection

Pugh charts used rotating datums to score each concept against 11 criteria including plastic removal effectiveness, wildlife safety, collection capacity, ease of maintenance, ease of deployment, durability and reliability, noise and visual intrusion, cost and ease of manufacturing, mobility and scalability, and risk of technical development (Appendix C). Aggregated net scores favored the Air Boat at +3.5 average across rotations, balancing mobility with low risk and outperforming the Drone at -1.5 and others.

3.5 Prototyping

To prove feasibility, prototyping was prioritized over heavy analysis. Two distinct prototyping phases were conducted across the fall and spring semesters: an early concept-validation phase focused on debris capture and propulsion approaches, and a hull-integration phase used to test buoyancy, layout, and subsystem placement before manufacturing the final SS Guardian. Together these prototypes provided critical feedback on geometry, performance, and operational feasibility, informing both concept

selection and detailed design refinement. Photos from both phases can be found in Appendix G.

3.5.1 Concept Validation Prototypes (Fall)

The fall semester focused on proving feasibility of the leading concepts before committing to a single design path. Two physical prototypes were built and tested:

- **Comb bulldozer mechanism:** A simple PVC frame fitted with paint stick "tines" was assembled to evaluate the feasibility of pushing or scooping floating debris. This prototype was tested in a brushy lake environment to assess how the comb interacted with surface vegetation and debris under realistic conditions. The test demonstrated that a tined geometry could effectively gather floating macroplastics while allowing water and thin vegetation to pass, validating the rake-style approach later adopted on the SS Guardian.
- **Foam-hull proof-of-concept airboat:** A small insulation foam hull was paired with a handheld fan sprayer to evaluate the basic feasibility of air propulsion on a foam vessel. Testing in a bathtub and swimming pool confirmed that an air-driven foam hull could maintain buoyancy and forward motion, supporting the decision to pursue an airboat architecture rather than a water-propelled design.

These early prototypes were intentionally rough but provided fast, low-cost answers to the most critical concept-level questions: whether tined debris capture worked in real water, and whether an air-propelled foam hull was a viable platform.

3.5.2 Test Boat and Hull V1 (Spring)

Following concept selection, a first-iteration test boat was built in the spring semester to validate the integrated system before committing to final manufacturing. This prototype, referred to as Hull v1, used the same XPS foam construction approach planned for the final boat but without the full fiberglass and epoxy laminate. Hull V1 served as a working platform for iterating on subsystem placement, propulsion configuration, and buoyancy under realistic loading conditions. Key activities performed on Hull V1 included:

- **Preliminary buoyancy testing:** The hull was loaded with mock and actual components to confirm that the planned three-layer foam construction could support the full weight of the system with adequate freeboard. This testing validated the 24 in × 12 in × 3 in hull dimensions before final manufacture.
- **Subsystem layout iteration:** The motor, propeller, electronics bay, battery, rudder, and rake mounting positions were repositioned and re-tested multiple times to optimize weight distribution, center of gravity, and accessibility for maintenance. Several layouts were trialed before settling on the final configuration.
- **Propulsion testing and failure mode discovery:** Initial testing on Hull V1 revealed several critical issues with the original propulsion setup, including a 3D-printed PLA propeller that melted off the motor shaft due to heat from the

brushed Traxxas 540 motor. This led to the selection of a 10-inch nylon propeller and a metal collet adapter for the final design.

- **Steering validation:** The transition from a water rudder to an air rudder was driven by observations on Hull V1, where water rudder authority was insufficient at the low operating speeds typical of debris collection in vegetation.

By the time Hull v1 testing was complete, every major subsystem had been validated or revised based on observed performance. This allowed the final hull to be manufactured with high confidence in the integrated design, minimizing rework on the more time-intensive fiberglass laminate construction.

4. Final Design

The final design of our air boat system was updated since the CDR to show a balance between performance, reliability, and maintainability in real lake environments. A defining feature of this concept is the placement of the thrust system to be entirely above the waterline which will reduce the risk of entanglements with underwater vegetation, a severe failure mode established in our FMEA. By relying on an air propulsion system, the boat can easily navigate through areas with dense plant growth and floating debris while maintaining directional control.

The propulsion system underwent significant changes. The original PLA propeller was replaced with an industry-standard glass-filled nylon propeller (2-blade, 10 in diameter $\times$ 7 in pitch) to substantially increase thrust, as pool testing with the PLA propeller revealed insufficient forward speed. In addition, the nylon propeller offers a higher melting temperature and improved structural strength compared to the previous PLA propeller. The motor mount, originally referred to as the spoiler, experienced thermal issues during testing. These were resolved by incorporating internal cooling channels and chamfers into the mount geometry, the latter of which also improved the press fit between the motor and the mount. The shroud, previously called the prop guard and featuring a waffle-pattern face, was redesigned with a smooth, open face to eliminate airflow disruption. In its final form, the shroud serves to focus propwash directly onto the air rudder, improving steering authority.

Steering is provided by a servo-actuated air rudder positioned in the propeller wash. The rudder was switched from a submerged water rudder to an above-waterline air rudder to avoid vegetation entanglement. Its height, ultimately reaching ~9 inches, was iteratively increased across design revisions to maximize the area exposed to propwash, improving responsiveness at low speeds. The servo-to-rudder hardware was also upgraded to a more secure fastener arrangement to prevent slippage under load.

The rake underwent complete redesign relative to earlier prototypes. It is now a single-piece part printed in TPU with a gyroid infill at 20%, replacing a previous two-piece PLA design. The tine length was adjusted, fillets were added at high-stress junctions to eliminate the stress concentrations responsible for fractures in earlier prototypes, and the overall width was set to 12 inches, made possible by access to a large Raise E2 3D printer. The TPU material provides several advantages: it acts as a compliant bumper against obstacles, flexes around submerged vegetation rather than snagging, resists fracture under impact and remains stiff enough to reliably funnel floating debris.

Engineering drawings for the final air boat assembly are provided on the following pages and in Appendix E. The project's bill of materials and budget are presented in Appendix F.

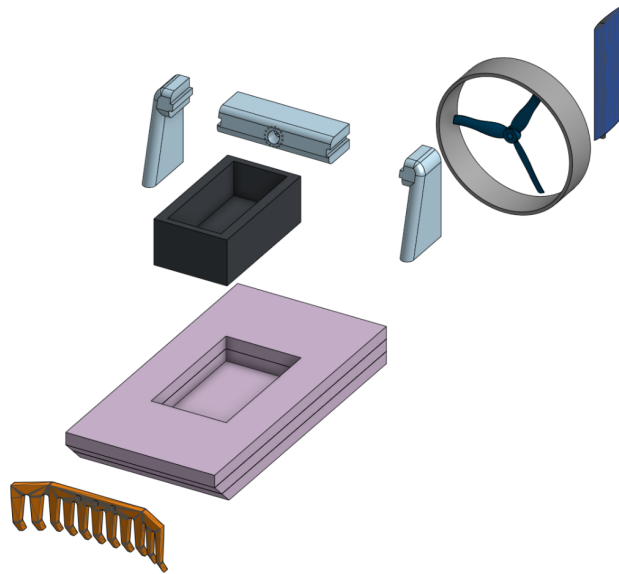


Figure 1. Exploded View of Airboat Assembly

5. Manufacturing and Testing

Having completed the critical design phase, the project moved into manufacturing and system verification. This section documents the full manufacturing and assembly process for the airboat, and the phased testing approach that guided the system from bench validation through field deployment on Lady Bird Lake. During this process, the team focused on creating solutions for any problems that arose during testing. By the end of the testing phase, the team had successfully deployed the airboat on Lady Bird Lake at the Keep Austin Beautiful Day.

5.1 Manufacturing and Assembly Plan

Manufacturing the final airboat was primarily done according to the process established in previous deliverables, with some slight modifications to include new pieces. The team employed the use of various hand tools and 3D printers throughout this process. The hull was constructed from three layers of XPS foam, each measuring 2 ft by 1 ft, manually cut using snap blades and fine-tooth saws. A hole was cut in the top two layers according to the finalized position of the electronics box. The bottom two layers received a 45-degree angle cut along the leading edge to create a draft for smoother water entry. The three layers were glued together with Gorilla Glue, using a ruler and square to ensure alignment, and then clamped overnight to cure. Once cured, all edges were rough-sanded and then fine-sanded to even out any discrepancies from cutting. After this, the entire hull was treated with a layer of epoxy, followed by fiberglass cloth, followed by a second layer of epoxy. After curing, the hull was finished with TotalBoat marine-grade paint for UV resistance and a professional appearance. Then finally, three holes were drilled through the front to the electronics box spaced according to the spacing of the rake screw holes.

Two mounting brackets were made with an 18-gauge aluminum sheet, cut to 3x1 in to act as a surface for the rake bolts to clamp onto. Like the hull, three holes were drilled into these mounting brackets according to the screw holes for the rake.

All remaining structural and mechanical components were produced via additive manufacturing on Raise 3D printers in TIW. The spoiler (motor housing) was redesigned to print into three pieces and glued together with superglue. The spoiler was printed at medium infill density with no supports. The rake was redesigned to print as one piece in the Raise E2 printer using TPU with a 20% infill density. The rake was printed at this infill, as it still allowed for sufficient flexing at the tips while maintaining rigidity at the thicker top portion. The servo mount, rudder, and propeller shroud were each simple single-piece prints requiring no supports. All printed parts were inspected for dimensional accuracy and surface quality before final assembly, and all were finished with black marine-grade paint.

Assembly proceeded in three stages. First, the spoiler, servo mount, and propeller shroud were glued onto the hull in their finalized positions. Second, the motor was press-fit into the spoiler and the propeller attached to the motor securely with the collet adapter. Third, the rake was secured to the hull with the use of three bolts, washers, nuts and the mounting brackets. The setup from front to rear was bolt head, rake, washer, mounting plate, hull to the center hole, mounting plate, and then the nut for all three bolts. The motor and servo were then wired through to the box, the rudder was mounted to the servo via a screw and nut, and the box was clamped shut for use. Electrical tape was used to secure any loose wires safely away from the propeller. The finalized airboat loaded mass was found to be 3kg after assembly.



Figure 2. Layering of Fiberglass over Hull during Construction

5.2 Testing Plan

Testing proceeded in three phases, progressing from controlled environments to realistic field conditions.

Phase 1 focused on electronics verification and basic buoyancy. Before assembly into the hull, the motor, ESC, servo, receiver, and battery underwent a bench test to confirm proper communication between the transmitter and receiver, correct motor spin direction and throttle response, and full servo range of motion for rudder actuation. From here, the buoyancy and balance of the finalized boat was confirmed through testing in shallow calm water at creeks and pools. The team was striving for a hull draft of less than 1 in and found it to be 0.5 in during testing.

Phase 2 focused on testing the redesigned steering system and TPU rake. The airboat was deployed in a calm creek to evaluate turning characteristics and performance of the TPU rake. The team intended for the turning radius to be tight at less than 0.5m. From this test it was found that the air rudder needed to be redesigned to a larger size to allow for more deflection of air, improving steering response and turn radius. The TPU rake was found to be durable under impact and strong enough to push trash found in the creek.

Phase 3 focused on debris collection performance in realistic conditions on Lady Bird Lake at Keep Austin Beautiful Day. The airboat was deployed in various vegetated shoreline areas where trash, like aluminum cans and plastic bottles, was found. The team piloted the airboat through these dense vegetation areas and successfully relocated the trash to an accessible pickup location. Furthermore, turning radius testing and top speed testing were performed with all final parts.

5.3 Current Progress

As of the FDR submission date, the team has completed all manufacturing, assembly, and testing goals that were set. A finalized prototype has been built and tested. Overall, the team is satisfied with the performance of the airboat and has highlighted key improvement areas.



Figure 3. Final SS Guardian at Lady Bird Lake

Phase 3 testing showcased the effectiveness of the TPU rake within these dense vegetation areas. The rake did not get caught on any water lettuce or other vegetation and was able to effectively capture trash on the front face of the boat. It was observed that low hanging brush would get caught in the propeller if the team was not careful with positioning the boat. This did not cause any damage to the propeller or other boat systems; however, it is still a concern. The airboat's steering performed well under calm conditions; however, when a wake was present, the boat became noticeably harder to control. Under calm conditions, it was found that the turning radius of the airboat was approximately 0.5 m achieving the team's goal. Furthermore, the airboat's top speed was found to be 1.2 m/s. Another issue discovered during testing was poor visibility from the shoreline in much denser areas. Key improvement areas identified from this

testing include brushless electronics, twin rudders for improved steering, a mesh shroud over the propeller, and an FPV camera for improved viewing while piloting the boat.

6. Summary

The SS Guardian successfully addresses the cleanup gap identified at the start of this project. On-water testing confirmed a hull draft under 1 inch, loaded mass of 3 kg, top speed of 1.2 powerem/s, and a 0.5 m turning radius. Propulsion analysis predicts 38.75 N of thrust against 12 N of cruise drag, a 3.25x margin that accommodates wind, weeds, and payload during shoreline operation. Deployment at Keep Austin Beautiful Day on April 18, 2026, demonstrated the system operating in its intended environment alongside volunteer cleanup crews.

The team spent \$188 against the \$250 budget. With foam reclaimed from a construction site and filament and electronics provided by the Cockrell School, total replication cost is approximately \$364, well within reach for a community group or small municipal partner. The most valuable lesson from the build was the cost of underspecifying material properties early on: PLA in the motor mount and rake, and a water-based rudder all failed in service and had to be redesigned. Each failure pushed the team toward materials and geometries better matched to their actual operating conditions.

Known limitations include no onboard storage (the rake pushes debris to shore, where a person must collect it), the need for a human operator, and the system's role as a complement to existing cleanup infrastructure rather than a replacement for systems like Mr. Trash Wheel. Future work includes a brushless drivetrain for more power and less heat, twin-rudder steering on a second servo, a fine-mesh shroud for lower-hanging brush, and an FPV camera for improved operator visibility. The platform is positioned for handoff to Watershed Protection or community partners interested in extending its use.

7. Acknowledgments

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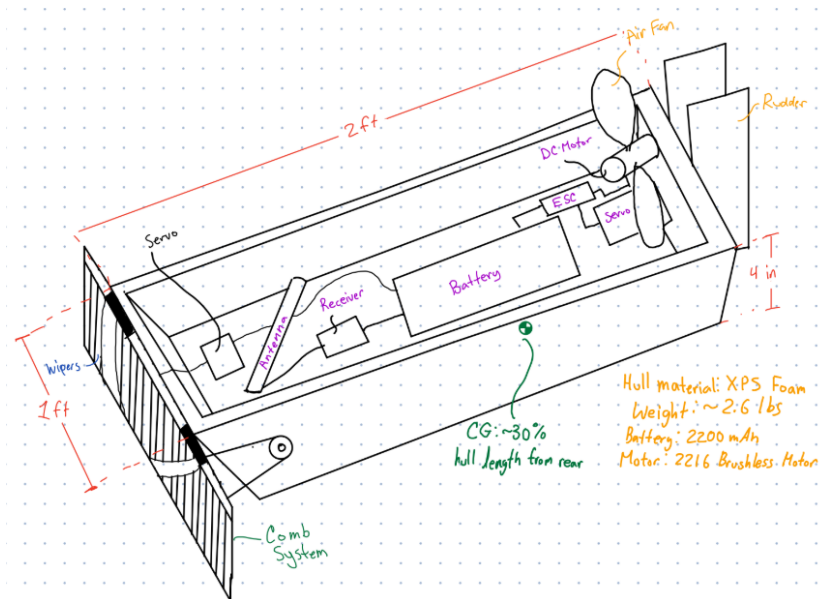
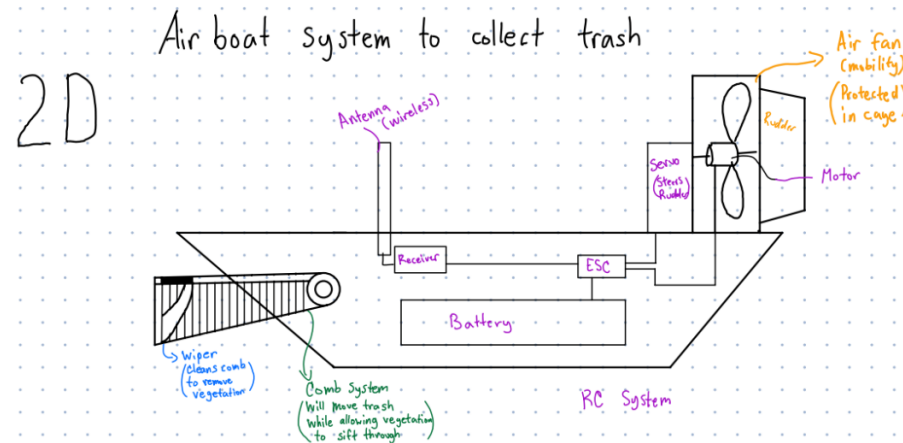
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Appendices

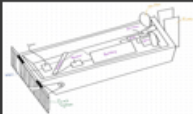
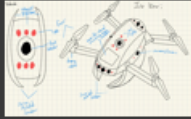
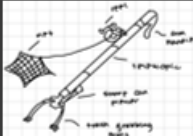
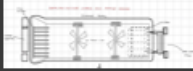
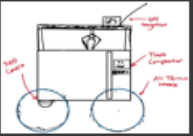
Appendix A Engineering Specifications

Dates	Imp.	Specification	Metric	Target Value	Verification Method	Responsibility	Source / Justification
26-Nov	W	Routine service burden (avg. time per maintenance visit)	Minutes / visit	≤ 20 min	Time study during maintenance	Team	Customer Needs
26-Nov	D	Installation / relocation burden	Two-person install time (h)	≤ 2 h	Setup trial	Team	Customer Needs
26-Nov	D	Single Module Mass	Mass per module	≤ 23 kg module	Weight test	Team	Customer Needs; NIOSH Lifting Equation
26-Nov	D	Project Budget	Project Budget	≤ \$250	Review of BoM	Team	Customer Needs / Project course guidelines
26-Nov	D	Visible plastic items per 100 m shoreline (standard 30-min pass)	Count	≤ 2 items / 100 meter	Field survey + video documentation	Team	Customer Needs
26-Nov	D	Floating macro-plastics (≥ 5 mm) per 50 m surface transect	Count	≤ 3 items / 50 meter	Transect sampling + visual inspection	Team	Customer Needs
26-Nov	D	Macro plastic mass removed per operating day	kg / day	≥ 1 kg/day	Field trial + weigh-scale validation	Team	(United Nations International Telecommunication Union, 2020)
26-Nov	D	Microplastic presence in weekly grab samples (≥ 300 µm sieve)	% of samples with detectable plastics	≤ 20 %	Lab filtration + microscopy	Team	Customer Needs; US EPA sediment/turbidity criteria snapshot
26-Nov	D	Retention of captured plastics under agitation	% retained after 2 min wake test	≥ 95 %	Simulated 2 minute wake test	Team	Customer Needs; ASTM F1523 – Selection of Booms
26-Nov	W	Sediment disturbance during operation(Δ TSS 10 m downstream–upstream)	mg/L TSS	≤ +5 mg/L	Turbidity probe testing	Team	Customer Needs
26-Nov	D	Wildlife-safety compliance score	Pass/Fail	Pass	Safety audit + visual inspection	Team	Customer Needs
26-Nov	W	Sound level increase at 10 m distance	dB above Ambient	≤ 5 dB	Sound meter test	Team	Customer Needs; Austin Outdoor Amplified Sound
26-Nov	D	Public visual acceptance rating	Likert 1–5 survey	≥ 4	Public intercept survey	Team	Customer Needs
26-Nov	D	Public data transparency latency	Hours delay	≤ 24 h	Dashboard update log	Team	Customer Needs

Appendix B Concept Sketches



Appendix C
Pugh Chart – Air Boat

Only use +, -, or S		Concepts				
		Air Boat (Datum)	Drone	Net Gun	Floating Channel	Creek Crawler
						
Criteria	Plastic Removal Effectiveness	0	-	0	+	0
	Wildlife Safety	0	0	0	0	+
	Public Transparency	0	0	-	0	0
	Collection Capacity	0	-	-	0	0
	Ease of Maintenance	0	-	+	0	0
	Ease of Deployment Relocation	0	+	+	+	0
	Durability and Reliability	0	0	0	0	0
	Noise and Visual Intrusion	0	0	+	0	0
	Cost and Ease of Manufacturing	0	-	+	0	-
	Mobility and Scability	0	+	+	0	0
	Risk of Technical Development	0	0	+	-	0
Totals	Positive Count	0+	2	5	2	1
	Negative Count	0-	4	2	0	1
	Total Sum	0	-2	3	2	0
	Rank	X	4	1	2	3

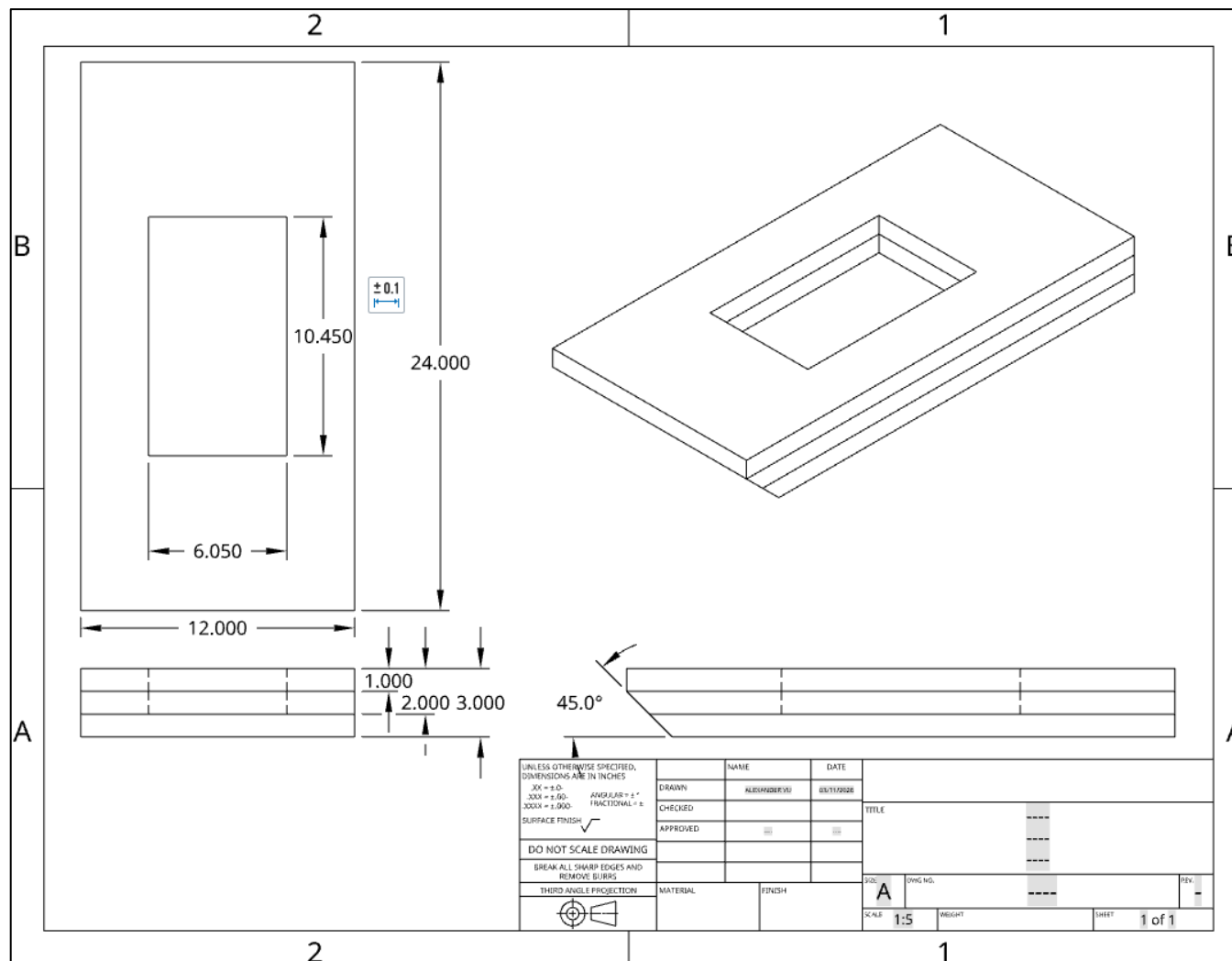
Appendix D

Failure Mode and Effects Analysis

				Current Situation					Improved Situation					
Failure Location	Failure Mode	Failure Effect	Failure Cause	Current Detection Steps	S	O	D	RPN	Suggested Remedial Measures	Revision	S	O	D	RPN
Floatation	Puncture	Partial/Full Sinking	Collision	Visual Inspection before launch	9	3	6	162	Add HDPE on bottom/bumper					
Floatation	Wear	Loss of buoyancy	Rough Handling	Visual Inspection before launch	9	3	6	162	Add HDPE on bottom/bumper					
Floatation	Flooding	Electronics swamped	Weight/Water Height	Monitor boat trim	8	4	5	160	Raise deck height/thickness					
Comb	Vegetation Jam	Boat stalls	Heavy vegetation	Watch comb for buildup	7	6	6	252	Angled weed-deflector bar	Angled weed-deflector bar	7	3	4	84
Comb	Buckling	Reduced Efficiency	Impacts	Visual Inspection before launch	6	4	6	144	HDPE comb					
Comb	Rotting	Reduced Efficiency	Water	Visual Inspection before launch	6	4	6	144	Laminate Wood/HDPE					
Comb	Trash Slippage	Wasted Effort	No side rails/steering	Experimental Testing	5	5	5	125	Add low perimeter rail/curvature					
Propulsion	Hit by Debris	Loss of thrust	No debris guard	Scanning Environment/Steering	9	2	4	72	Mesh Fan Guard					
Propulsion	Loose Fitting	Loss of control	Insufficient fasteners	Physical Test before launch	6	5	5	150	Lock nuts/Threads					
Electronics	Flooding	Power Loss	Splashing	Check box for leaks/condensatino	9	4	5	180	Electronics enclosure	Electronics enclosure	9	2	3	54
Electronics	Loose Wires	Power Loss	Vibration	Monitor steering responses	7	4	5	140	Locking connectors/clamps					
Electronics	Signal Loss	Loss of control	Range/Battery	Perform range test	8	5	6	240	Receiver failsafe/high antenna	Receiver failsafe/high antenna	8	3	4	96

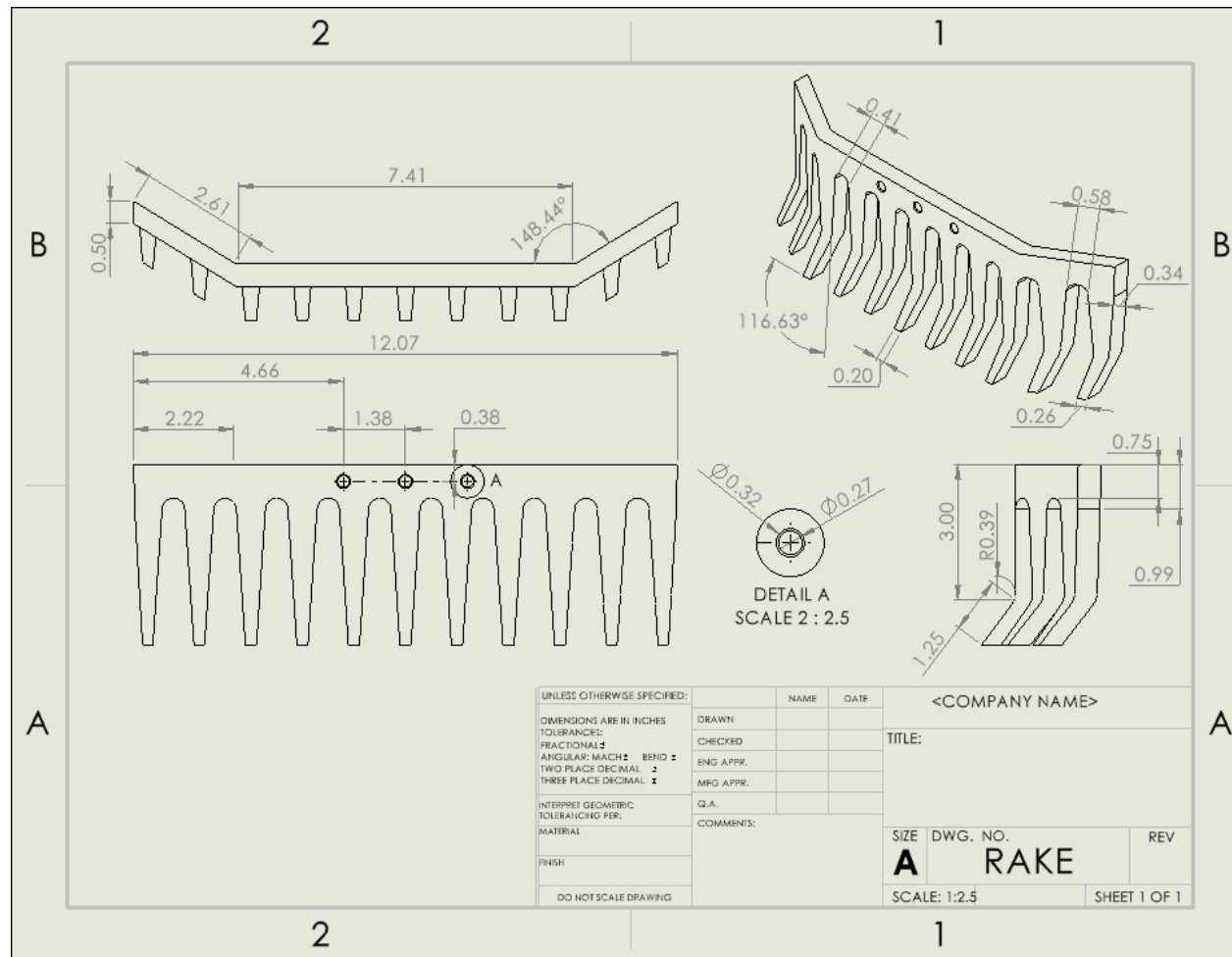
Appendix E

Engineering Drawings – Hull



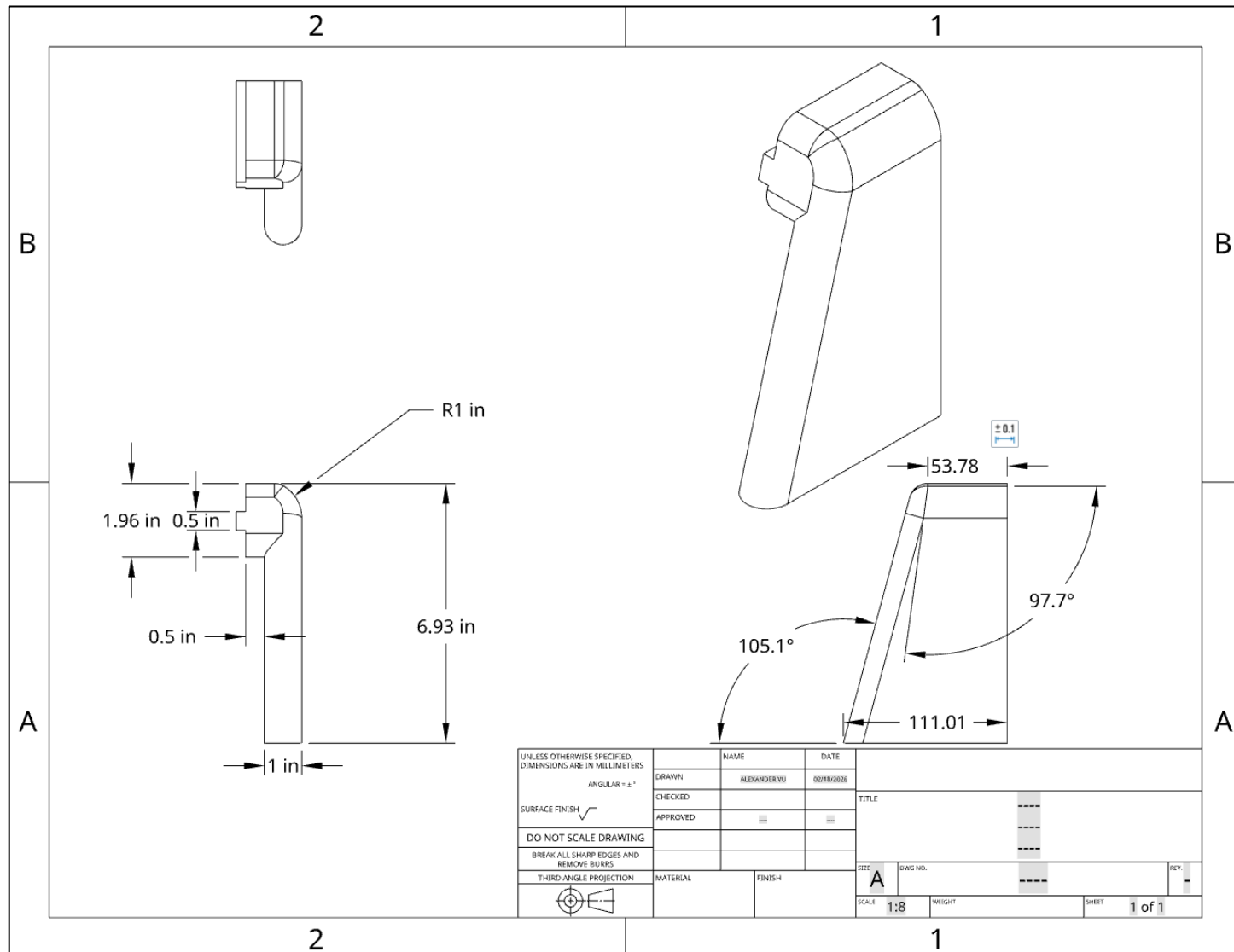
Appendix E

Engineering Drawings – Rake



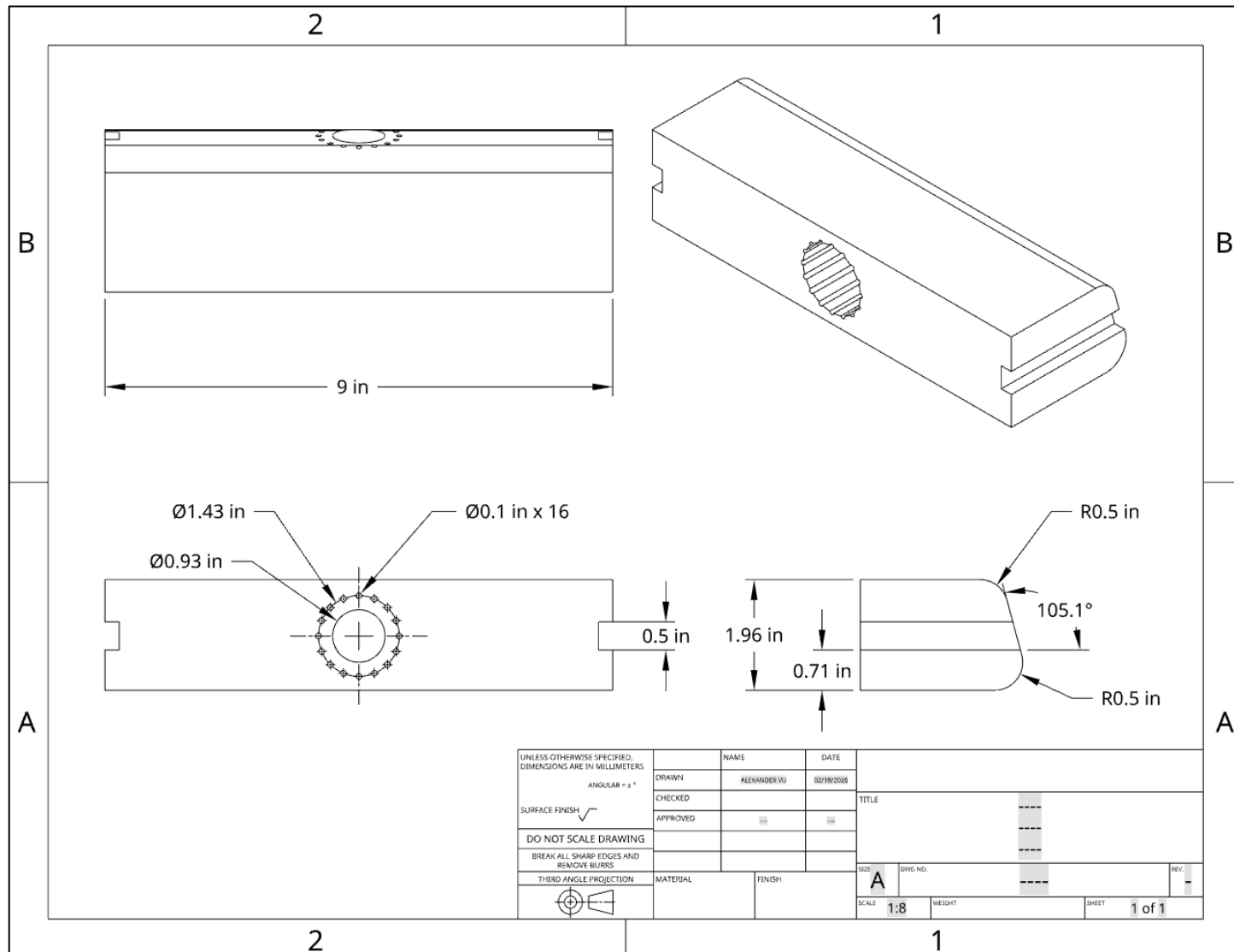
Appendix E

Engineering Drawings – Spoiler Leg



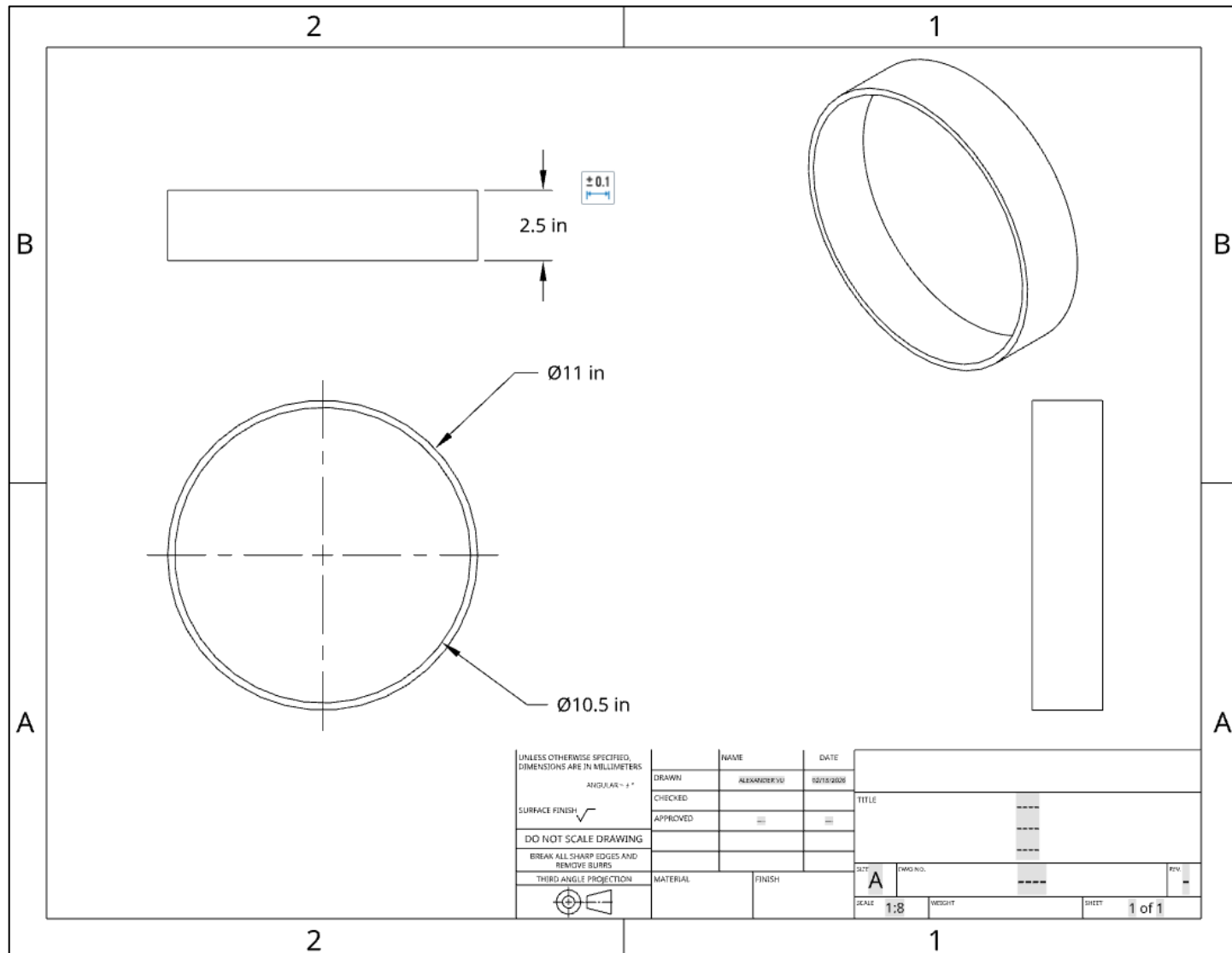
Appendix E

Engineering Drawings – Spoiler Top



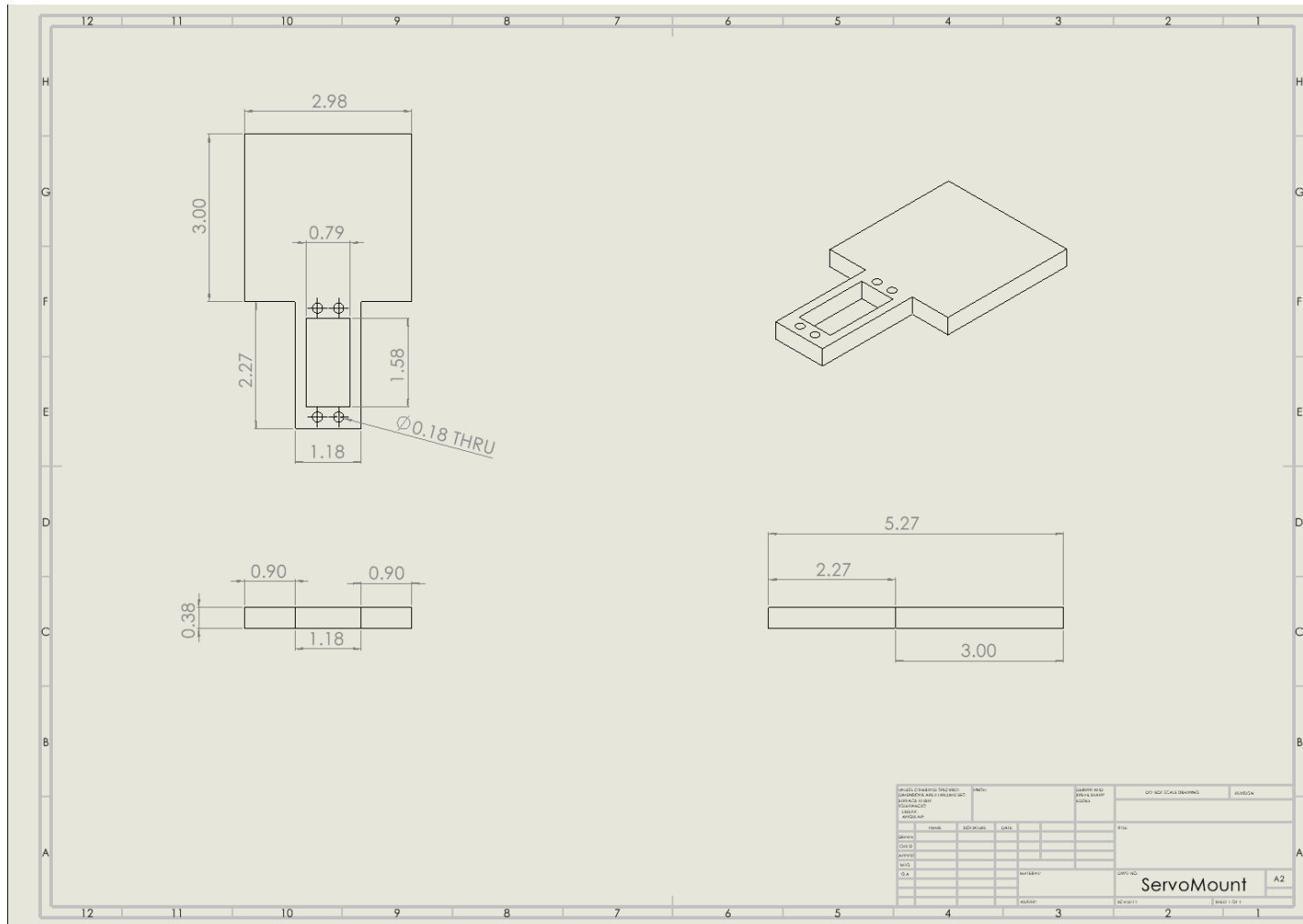
Appendix E

Engineering Drawings – Shroud



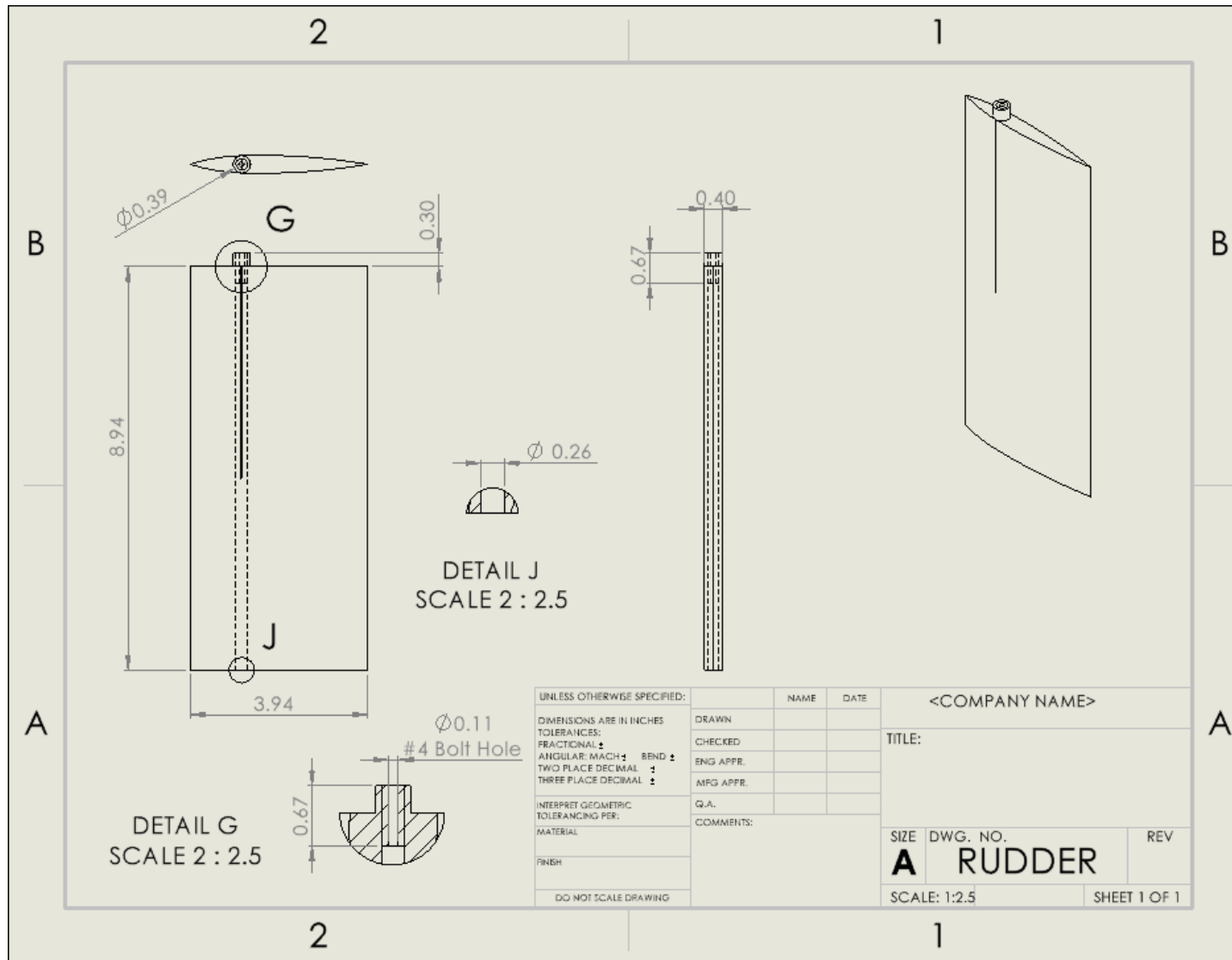
Appendix E

Engineering Drawings – Servo Holder



Appendix E

Engineering Drawings– Rudder



Appendix F

Bill of Materials and Budget

Item	Description	Quantity	Total Price	Source	Notes/Justification
Fiberglass Cloth	6 oz, 50" wide, 6 feet	1	\$9.99	Amazon	Reinforcement for durability; adds tensile strength (45,000 psi when bonded); compatible with epoxy wet-out for waterproofing.
TotalBoat Epoxy Resin	Epoxy resin, 2 quarts	1	\$35.60	Amazon	Epoxy for joining, sealing, and fiberglass; non-toxic, water-safe; cures waterproof for freshwater use.
Marine Paint	TotalBoat Topcoat, Aqua Mist	1	\$52.00	Amazon	Final aesthetic layer; UV-resistant, professional finish; covers 80-100 sq ft.
Outside Electrical Box	Waterproof box for electronics	1	\$9.99	Amazon	Waterproof box for electronics
Various System Components	Rake, rudder, servo mount, fan, fan guard, spoiler	variable	\$0.00	TIW	Custom 3D printed components with provided free filament from Cockrell School
Bolts	1/4 in.-20 x 6 in. Zinc Plated Hex Bolt	6	\$3.84	Home Depot	Component for attaching rake to foam
Nuts	Marine Grade Stainless Steel 1/4-20 Hex Nut (4 Pieces)	2	\$4.98	Home Depot	Component for attaching rake to foam
Black Spray Paint	12 oz. Flat Black General Purpose Spray Paint	2	\$13.96	Home Depot	Aesthetic layer for 3D printed components
Clear Coat Spray Paint	12 oz. Gloss Clear General Purpose Spray Paint	1	\$6.98	Home Depot	Aesthetic layer for 3D printed components
Propellor	uxcell RC Propellers CW 8x6 Inch 2-Vane	1	\$15.59	Amazon	Propellor for thrust and power
Collet Adapters	6 Pack RC Propeller Adapter, 3.17mm Motor Shaft Aperture to 6mm Output Axle	1	\$9.99	Amazon	Physical connection for prop and motor
Gorilla Glue	Gorilla Clear Glue, 3.75 Ounce Bottle (Pack of 1) - Crystal Clear, All Purpose, and Water Resistant	1	\$9.48	Amazon	Adhesive for 3d printed parts
Rake TPU prints	2x Custom printed rakes at TIW made out of TPU filament	1	\$16.08	TIW	Custom printed rakes at TIW made out of TPU filament

Spend to Date	\$188.48
Total Budget	\$250.00
Budget Remaining	\$61.52
Percentage Utilized	75.39%

Appendix G
Prototyping Photos



Appendix G
Prototyping Photo



Appendix H
On-Water Testing Photo



Appendix I

Test Results Summary – Engineering Specs

#	Specification	Target	Result	Status
1	Routine service burden	≤ 20 min	~10 min between deployments (component inspection, battery swap, rake clean)	Met (Measured)
2	Installation / relocation burden	≤ 2 h	< 15 min by two people (boat fits in a standard car)	Met (Measured)
3	Single module mass	≤ 23 kg	3 kg loaded mass	Met (Measured)
4	Project budget	≤ \$250	\$188 spent	Met (Measured)
5	Visible plastic items per 100 m shoreline (post-pass)	≤ 2 items / 100 m	Target achieved during Keep Austin Beautiful Day deployment	Met (Measured)
6	Floating macroplastics per 50 m surface transect	≤ 3 items / 50 m	Target achieved during Keep Austin Beautiful Day deployment	Met (Measured)
13	Public visual acceptance	≥ 4 / 5	Polled attendees at Keep Austin Beautiful Day; all respondents rated the SS Guardian 5/5	Met (Measured)
7	Macro plastic mass removed per operating day	≥ 1 kg/day	Substantial debris volume collected during a 90-minute deployment at Keep Austin Beautiful Day, supporting a projected daily throughput well in excess of the 1 kg/day target	Met (Inferred)
9	Retention of captured plastics under agitation	≥ 95%	Not formally tested with ASTM F1523 wake test; rake geometry observed to retain captured debris during turning and operation	Met (Inferred)
10	Sediment disturbance during operation (Δ TSS)	≤ +5 mg/L	Air propulsion and < 1-inch hull draft minimize bottom interaction; no visible sediment plume during testing	Met (Inferred)
11	Wildlife-safety compliance	Pass	No prop-water contact, prop guard installed, operating speeds well below EPA 0.3 m/s intake guideline at the rake	Met (Inferred)
12	Sound level increase at 10 m	≤ 5 dB	Subjectively quiet during operation; at 10 m distance, sound increase is well below the 5 dB threshold and comparable to ambient lakeside conditions	Met (Inferred)
8	Microplastic presence in weekly grab samples	≤ 20%	Microplastic remediation deferred in favor of macroplastic interception; outside scope of final build	Out of Scope
14	Public data transparency latency	≤ 24 h	Intended for a future deployed-fleet implementation; not applicable to current build	Out of Scope

Summary: Of 14 customer-needs specifications, 7 were directly measured and met, 5 were inferred met based on design choices and observed performance, and 2 were out of scope for the final build. All specifications relevant to the SS Guardian's macroplastic-collection mission were met or exceeded.

Appendix I

Test Results Summary – Technical Performance

Parameter	Predicted / Target	Result	Status
Thrust-to-drag margin	$\geq 2\times$	3.25 $\times$ predicted margin, confirmed by ability to push through vegetation with throttle headroom remaining	Met (Measured)
Top speed	≥ 1.0 m/s	1.2 m/s measured	Met (Measured)
Turning radius	≤ 1.0 m	0.5 m measured	Met (Measured)
Hull draft (loaded)	≤ 1.0 in	0.5 in measured	Met (Measured)
Loaded system mass	≤ 5 kg	3 kg measured	Met (Measured)
Propulsion temperature stability	No thermal failures	Final 10 $\times$ 7 nylon prop with collet adapter operated without thermal issues; replaced PLA prop that melted on Hull V1	Met (Measured)
Steering authority at low speed	Responsive in vegetation	Air rudder in prop wash provided steering authority at all throttle settings, including stationary	Met (Measured)
Hull water resistance	No water ingress to electronics bay	Fiberglass and epoxy laminate plus sealed electronics box maintained dry interior throughout testing	Met (Measured)
Replication cost	$\leq$ \$500 (target for accessibility)	\$364 total replication cost	Met (Measured)
Buoyant capacity	≥ 10 lbs (design margin)	~31 lbs theoretical maximum; operates well below limit	Met (Inferred)
Runtime per battery charge	≥ 20 min	20–30 min estimated based on 7.2 V 3000 mAh NiMH battery and ~7.5 A average system draw	Met (Inferred)

Summary: Of 11 technical performance parameters, 9 were directly measured and met, and 2 were inferred met based on operational behavior and design analysis. All targets were achieved, with several measured values significantly exceeding their requirements.